



Comparing hand-crafted and deep learning approaches for detecting AI-generated text: performance, generalization, and linguistic insights

Ramtin Ardeshtirifar¹

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Abstract

This study investigates techniques for detecting machine-generated text, a critical task in the era of advanced language models. We compare two approaches: a hand-crafted feature-based method and a deep learning method using RoBERTa. Experiments were conducted on diverse datasets, including the Human ChatGPT Comparison Corpus (HC3) and GPT-2 outputs. The hand-crafted approach achieved 94% F1 score on HC3 but struggled with cross-dataset generalization. In contrast, the RoBERTa-based method demonstrated superior performance and adaptability, achieving 98% F1 score on HC3 and 97.68% on GPT-2. Our findings underscore the need for adaptive detection methods as language models evolve. This research contributes to the development of robust techniques for identifying AI-generated content, addressing critical challenges in AI ethics and responsible technology use.

Keywords Machine-generated text detection · Natural Language processing · Artificial intelligence ethics · RoBERTa

1 Introduction

The rapid advancement of large language models (LLMs) has revolutionized natural language processing, enabling unprecedented capabilities in text generation. While these developments offer immense potential for beneficial applications, they also present significant challenges in detecting machine-generated content. The ability to generate highly coherent and contextually appropriate text raises concerns about potential misuse, from spreading disinformation to sophisticated phishing attacks [1, 2].

Recent research has highlighted both the sophistication of modern language models and the difficulty humans face in identifying their outputs. Spitale et al. [3] demonstrated that humans struggle to reliably distinguish between human-written and GPT-3 generated content, achieving accuracy rates only slightly above random chance. This challenge is compounded by the emergence of malicious applications, as evidenced by tools like WormGPT [4] and FraudGPT [5], which specifically target harmful use cases.

The field of machine-generated text detection has evolved to address these challenges, with approaches ranging from statistical analysis to deep learning methods. While early detection systems relied on basic linguistic features [6], modern approaches increasingly leverage transformer-based architectures and sophisticated feature engineering. However, current detection methods often struggle with crossmodel generalization and adaptability to new language models [7].

This study advances the field by comparing two distinct approaches: a traditional method based on handcrafted linguistic features and a deep learning approach using RoBERTa. Our research specifically addresses three critical questions: (1) How effective are hand-crafted linguistic features in distinguishing between human and machine-generated text? (2) How does the performance of deep learning models compare to traditional methods? (3) To what extent can these approaches generalize across different datasets and language models?

To explore these questions, we conduct extensive experiments using diverse datasets, including the Human ChatGPT Comparison Corpus (HC3) [8] and GPT-2 outputs

✉ Ramtin Ardeshtirifar
r.ardeshirifar@sussex.ac.uk

¹ Department of Informatics, University of Sussex, Sussex, UK

[9]. Our work contributes to the field by providing empirical insights into the effectiveness of different detection approaches, introducing novel linguistic features for detection, and offering practical recommendations for developing robust detection systems. The remainder of this paper is organized as follows: Sect. 2 reviews related work, Sect. 3 details our methodology, Sect. 4 presents our results, and Sect. 5 discusses implications and future directions.

2 Related work

Early research in machine-generated text detection primarily focused on identifying content from basic language models and machine translation systems. For instance, Nguyen-Son et al. [10] employed statistical analysis of word pair frequencies, while Badaskar et al. [6] explored basic linguistic features for detection. However, the landscape has evolved significantly with the advent of more sophisticated language models.

The release of GPT-2 marked a pivotal moment in detection research. Solaiman et al. [1] demonstrated that a simple classifier using TF-IDF features could achieve 88% accuracy on outputs from the 124 M parameter model, though performance decreased to 74% for the 1.5B parameter version. Zellers et al. [11] advanced the field by developing the Grover model which could detect its own outputs with 92% accuracy, demonstrating the potential of using the generative model itself for detection purposes. Recent work has focused on more sophisticated detection approaches. Mitchell et al. [12] introduced DetectGPT, a zero-shot detection method that leverages probability curvature to identify machine-generated text without requiring training data.

In the context of more recent language models, Guo et al. [8] developed the HC3 dataset specifically for comparing human and ChatGPT outputs. Their analysis revealed distinct patterns in lexical diversity and syntactic structure between human and machine-generated text. Similarly, Pu et al. [7] conducted a comprehensive study of detection limitations and opportunities, highlighting challenges in detector generalization across different language models and domains.

The emergence of RoBERTa-based approaches, building on Liu et al. [13]’s work, has shown particular promise. Yu et al. [14] introduced GPT Paternity Test (GPTPat), which leverages ChatGPT to generate questions and re-answers for text under scrutiny, comparing similarities between original and re-answered text to determine if content is machine-generated. Their approach achieved 94.57% accuracy on generalization test sets and showed particular robustness against re-translation and polishing attacks.

Recent studies have also highlighted the human factor in detection. Spitale et al. [3] found that humans struggle to reliably distinguish between human-written and GPT-3 generated content, with accuracy rates only slightly above random chance. This underscores the importance of developing automated detection systems.

The rising concern about malicious applications of language models, exemplified by the emergence of WormGPT [4] and FraudGPT [5], has added urgency to detection research. These developments highlight the need for robust detection methods that can adapt to evolving threats while maintaining high accuracy across different models and domains.

Our work builds upon these foundations by providing a systematic comparison between hand-crafted feature-based and deep learning approaches, with particular attention to cross-model generalization and the effectiveness of linguistic features identified in previous studies. We extend existing work by incorporating advanced metrics like the Fano factor [15] and comprehensive dependency parsing statistics, while also leveraging the power of transformer-based architectures for detection.

3 Methods

This study employed two distinct approaches to detect machine-generated text: a hand-crafted feature-based method and a deep learning-based approach using RoBERTa. We evaluated these methods on multiple datasets to assess their effectiveness and generalizability.

3.1 Datasets

We utilized two primary datasets, as summarized in Table 1.

The HC3 dataset was chosen for its comprehensive comparison between human and AI-generated responses across various domains. It includes both versions (with and without indicating words) to assess the impact of these linguistic cues on detection performance. The GPT-2 Output Dataset was selected to evaluate our methods on text generated by a different language model, testing cross-model generalization.

Table 1 Summary of datasets used in the study

Dataset	Description	Key Features
Human ChatGPT Comparison	24,322 questions 58,546 human responses	Two versions: with and without indicating words
Corpus (HC3)	26,903 ChatGPT responses	
GPT-2 Output Dataset	250,000 samples from GPT-2 medium (355 M parameters)	Samples vary from 5 to 1024 tokens (avg. ~580)

3.2 Data preprocessing

We applied a rigorous preprocessing pipeline to ensure data consistency and remove potential biases:

1. Punctuation and contraction correction: We standardized punctuation usage and ensured proper formatting of contractions to eliminate superficial differences between human and AI-generated text.
2. Removal of extraneous elements: We eliminated asterisks, web-related text (URLs, mentions), and HTML entities to focus on core linguistic content.
3. Text normalization: Spaces were normalized and Unicode characters removed to ensure consistency across all text samples.
4. Length filtering: Texts with fewer than 10 tokens were excluded to ensure sufficient content for meaningful analysis.

For the GPT-2 dataset, we randomly sampled 25,000 entries each for human and machine-generated text due to computational constraints. This balanced subset allowed for fair comparison while managing processing requirements.

3.3 Hand-crafted feature-based approach

Our hand-crafted approach leveraged a diverse set of linguistic features, as summarized in Table 2.

3.3.1 Feature extraction and normalization

We extracted a variety of linguistic features to capture different aspects of text complexity and style. These features can be broadly categorized into lexical, syntactic, and statistical measures.

Lexical features Lexical diversity was calculated using the type-token ratio, providing a measure of vocabulary richness. This metric quantifies the proportion of unique words (types) to the total number of words (tokens) in a given text.

Syntactic features We used the spaCy library [16] for part-of-speech (POS) tagging and dependency parsing. From

these analyses, we extracted the frequency of each POS tag and the top 30 most frequent dependency relations. These features provide insights into the grammatical structure and complexity of the text.

Statistical features Perplexity, computed using the GPT-2 model, was used to quantify the predictability of the text. Lower perplexity scores indicate that the text is more predictable given the language model, while higher scores suggest greater variability and unpredictability.

Additionally, we incorporated the Fano factor [15] as a measure of dispersion in word frequencies. Historically rooted in information theory and signal processing, the Fano factor quantifies the variability of a probability distribution. In the context of text analysis, it captures the variability in word usage across the text. The Fano factor was computed as follows:

$$F = \frac{\sigma^2}{\mu} \quad (1)$$

Where:

- F is the Fano factor value.
- σ^2 represents the variance of word frequencies in the text.
- μ represents the mean of word frequencies in the text.

Normalization To ensure fair comparison across texts of varying lengths, we applied normalization to the extracted features. This was crucial because raw frequencies of features like POS tags or dependency relations can be influenced by the length of the text. We used the following normalization approach.

$$\text{Normalized Frequency} = \frac{\text{Raw Frequency}}{\text{Total Word Count}} \quad (2)$$

This normalization was applied to lexical diversity, POS tag frequencies, dependency relation frequencies, and the Fano factor. By dividing the raw frequency of each feature by the total word count, we obtained a normalized frequency that represents the feature's prevalence relative to the text's length.

3.3.2 Classification

For classification, we employed the XGBoost algorithm, known for its efficiency and performance in handling structured data [17]. We used default parameters and the model

Table 2 Features used in the hand-crafted approach

Feature category	Description
Lexical diversity	Ratio of unique words to total words
Perplexity	Mean and standard deviation
Syntactic features	POS tagging, top 30 dependency relations
Sentence structure	Mean and SD of sentence lengths, Average clauses per sentence
Fano factor	Variance of word frequencies divided by the mean

Table 3 RoBERTa-based model architecture and training configuration

Component	Description
Base model	RoBERTa for contextual embeddings
Fully connected layers	Layer 1: 768 to 32 dimensions Layer 2: 32 to 1 dimension
Activation functions	ReLU between FC layers, Sigmoid for output
Tokenization	RoBERTa's native tokenizer
Sequence length	Truncated or padded to 500 tokens
Batch size	6
Epochs	1
Learning rate	1×10^{-5}

was trained on 90% of the data, with 10% reserved for validation to monitor generalization during training.

3.4 Deep learning-based approach

Our deep learning approach utilized RoBERTa, a robustly optimized BERT model [13]. The architecture and training configuration are presented in Table 3.

3.4.1 Model architecture

We fine-tuned the RoBERTa base model, leveraging its pre-trained knowledge of language patterns. The model's output was fed into two fully connected layers, reducing dimensionality from 768 to 32, and then to a single output neuron. ReLU activation was used between the fully connected layers to introduce non-linearity, with a final sigmoid activation for binary classification.

3.4.2 Training process

Texts were tokenized using RoBERTa's native tokenizer and padded or truncated to a fixed length of 500 tokens, balancing information retention with computational efficiency. We employed a small batch size of 6 and a low learning rate of 1×10^{-5} to fine-tune the pre-trained model effectively. Training was limited to one epoch to prevent overfitting, given the model's strong pre-trained foundation.

3.5 Model evaluation and comparative analysis

Both approaches were evaluated using the original test data provided with the datasets, ensuring consistency with previous studies. We used the classification report function from scikit-learn [18] to compute precision, recall, and F1 scores.

Our comparative analysis encompassed:

Table 4 F1 scores (%) for Hand-crafted Feature-based approach

Trained on ↓ Tested on →	HC3-with	HC3-without	GPT-2
HC3 with indicating words	94	93	20
HC3 without indicating words	94	94	22
GPT-2	57	56	82

1. Cross-dataset performance: We trained models on one dataset and tested on another to assess generalization across different text sources and AI models.
2. Feature importance visualization: For the XGBoost model, we analyzed feature importance to understand which linguistic characteristics were most discriminative.
3. Linguistic feature distribution analysis: We compared the distribution of key features (e.g., lexical diversity, sentence length, perplexity) between human and AI-generated texts.
4. Syntactic pattern examination: We analyzed the distribution of POS tags and dependency relations to identify characteristic patterns in AI-generated text.

4 Results

Our study compared the performance of hand-crafted feature-based and deep learning approaches in detecting machine-generated text across multiple datasets. The results reveal significant insights into the effectiveness and generalizability of each method.

4.1 Hand-crafted feature-based approach

The XGBoost classifier, trained on hand-crafted features, demonstrated strong performance within individual datasets but showed limitations in cross-dataset generalization. Table 4 presents the F1 scores for this approach.

The model achieved a 94% F1 score when trained and tested on the HC3 dataset, both with and without indicating words. However, its performance dropped significantly (20–22% F1 score) when applied to the GPT-2 dataset, indicating poor generalization to different language models.

Feature importance analysis (see Appendix Figs.) revealed that the top three most important features for the HC3 dataset were **lexical diversity**, **average sentence length**, and **average perplexity**. For the GPT-2 dataset, the top three features were **lexical diversity**, **Fano factor**, and **possessive pronouns**.

4.2 Deep learning-based approach using RoBERTa

The RoBERTa-based model demonstrated superior performance and better generalization across datasets. Table 5 presents the F1 scores for this approach.

The model achieved an impressive 98% F1 score on the HC3 dataset and maintained strong performance (63.33% F1 score) when applied to the GPT-2 dataset, showcasing better cross-model generalization compared to the hand-crafted approach.

4.3 Linguistic pattern analysis

Examination of Part-of-Speech (POS) distributions revealed distinct patterns between human and machine-generated texts, as shown in Fig. 1.

Key observations from our analysis reveal distinct patterns between machinegenerated and human-written texts. Machine-generated texts consistently showed higher frequencies of nouns, determiners, adjectives, and auxiliaries. In contrast, human-written texts exhibited more frequent use of punctuation and adverbs. These differences in part-of-speech usage provide valuable insights into the linguistic characteristics of AI-generated content. Further examination through dependency relation analysis uncovered additional disparities. Machine-generated text demonstrated lower rates of certain relations compared to human-authored content. Specifically, we observed reduced occurrences of punctuations, nominal subjects, adverbial modifiers, and roots in AI-generated text. These findings suggest that current language models may struggle to fully replicate the complex syntactic structures commonly found in human writing.

4.4 Statistical patterns in human and Machine-Generated text

Figure 2 presents four key statistical distributions that differentiate human-authored from machine-generated texts.

Table 5 F1 scores (%) for deep Learning-based approach

Trained on ↓ Tested on →	HC3-with	HC3-without	GPT-2
HC3 with indicating words	98.00	97.69	63.33
HC3 without indicating words	98.00	97.68	63.33
GPT-2	65.49	65.59	97.68

Each distribution reveals distinct patterns in how language models generate text compared to human writing. Our analysis reveals several distinctive patterns:

- **Fano Factor Distribution** (Fig. 2a): Human-authored texts show a pronounced peak at lower Fano factor values (around 1–2), indicating more natural variation in word usage. Machine-generated texts exhibit a broader, flatter distribution extending to higher values, suggesting more irregular patterns in word frequency variation.
- **Sentence Length Variation** (Fig. 2b): Human writers demonstrate greater variability in sentence length, with a wider distribution and higher standard deviation. Machine-generated text shows a more concentrated distribution around shorter sentence lengths, indicating a tendency toward more uniform sentence structures.
- **Lexical Diversity** (Fig. 2c): Human-authored texts display a broader distribution of lexical diversity scores, centered around higher values (0.6–0.7), indicating richer vocabulary usage. Machine-generated texts cluster around lower values (0.5–0.6) with a narrower distribution, suggesting more conservative vocabulary choices.
- **Perplexity Patterns** (Fig. 2d): Human writing exhibits consistently higher perplexity scores with a broader distribution, reflecting greater unpredictability and creativity in language use. Machine-generated text shows lower perplexity values with a sharper peak, indicating more predictable patterns in word choice and sentence construction.

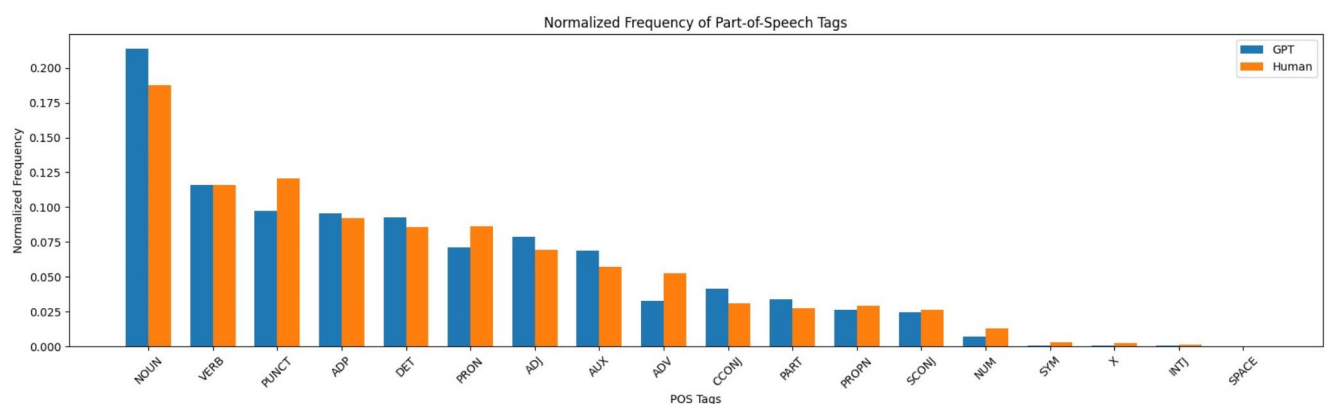
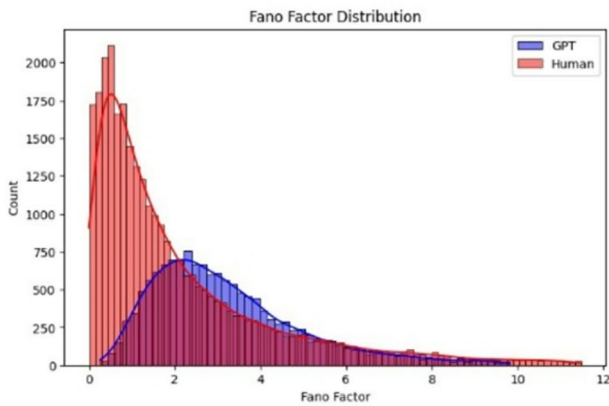
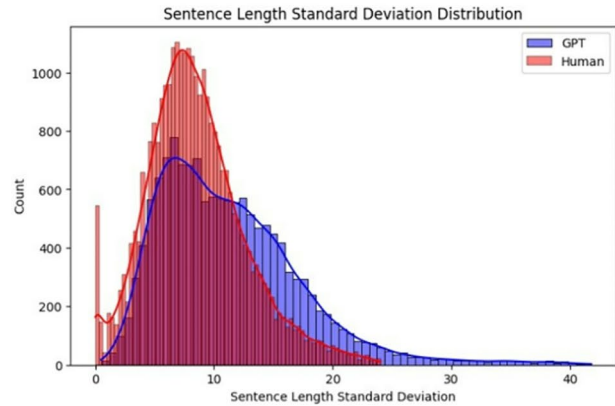


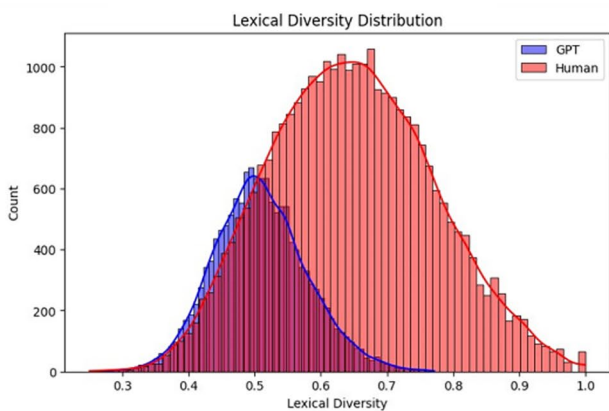
Fig. 1 Distribution of Part-of-Speech tags in human and machine-generated texts



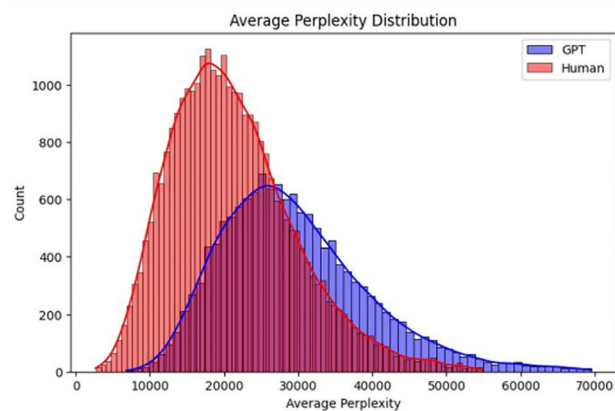
(a) Distribution of Fano Factor, showing how word frequency variations differ between human and machine text



(b) Distribution of sentence length standard deviation, illustrating structural variation in text



(c) Distribution of lexical diversity, measuring vocabulary richness



(d) Distribution of perplexity, indicating predictability of text sequences

Fig. 2 Statistical characteristics distinguishing human-authored and machinegenerated texts. Human writing consistently shows greater variability across all measures, while machine-generated text exhibits more constrained patterns

These distributions collectively demonstrate that while machine-generated text has achieved remarkable fluency, it still exhibits more constrained and predictable patterns across multiple linguistic dimensions compared to human writing. The consistent differences in these statistical measures provide valuable features for detection systems and insight into the current limitations of language models.

4.5 Cross-dataset performance

Both approaches struggled with cross-dataset generalization, particularly when trained on HC3 and tested on GPT-2. However, the RoBERTa-based model showed better adaptability:

- Hand-crafted approach: F1 score dropped from 94% to 20–22%.

- RoBERTa-based approach: F1 score dropped from 98 to 63.33%.

This performance gap highlights the challenge of detecting AI-generated text across different language models and domains.

In summary, while both approaches showed high performance within individual datasets, the deep learning method demonstrated superior generalization and adaptability. These results underscore the potential of transformer-based models in developing robust detection systems for AI-generated content.

5 Discussion

Our study provides a comprehensive comparison of hand-crafted feature-based and deep learning approaches for detecting machine-generated text. The findings offer significant insights into the effectiveness of these methods and their implications for AI ethics and text analysis.

5.1 Performance and generalization

The superior performance of our RoBERTa-based model (98% F1 score on HC3) represents a significant improvement over previous detection approaches. For comparison, Mitchell et al. [12]’s perplexity-based method achieved 85.6% accuracy on GPT-2 outputs, while Mitrović et al. [19]’s DistilBERT approach reached 93.2% F1 score on ChatGPT data. Our RoBERTa implementation’s 98% F1 score on HC3 and 97.68% on GPT-2 demonstrates meaningful progress in detection accuracy.

However, the performance degradation we observed in cross-dataset evaluations (98–63.33% F1 score when applying the RoBERTa model trained on HC3 to GPT-2 data) aligns with and quantifies a challenge noted but not thoroughly measured in previous work. While Zellers et al. [11] observed qualitatively that their GPT-2 detector struggled with newer models, our study provides concrete metrics for this degradation across different model generations. Similarly, while Guo et al. [8] noted detection challenges across different versions of ChatGPT, our work extends this analysis to cross-model detection between fundamentally different architectures (ChatGPT and GPT-2).

This generalization gap can be attributed to several factors that build upon previous findings:

1. **Architectural differences:** Expanding on Jawahar et al. [20]’s work on model architectures, our RESULTS demonstrate how different underlying architectures (GPT2 vs. ChatGPT) produce distinct linguistic signatures that challenge detection systems. While their work focused on architectural differences in generation quality, our study quantifies how these differences impact detection performance.
2. **Training data variation:** Our findings complement Solaiman et al. [1]’s work on dataset influence by showing how models trained on different corpora develop unique statistical patterns. Our cross-dataset experiments provide empirical evidence for this phenomenon’s impact on detection accuracy.

3. **Model evolution:** Building on Zellers et al. [11]’s observations about model evolution, our study provides the first quantitative analysis of how detection performance degrades across model generations. The significant drop in F1 scores (34.67% points) when transitioning from HC3 to GPT-2 detection offers a concrete metric for this challenge.

Our hand-crafted feature approach, while less successful in cross-dataset scenarios, provides novel insights into specific linguistic features that distinguish AI-generated text. Unlike Nguyen-Son et al. [10]’s focus on basic statistical measures, our comprehensive feature set includes advanced metrics like the Fano factor and dependency parsing statistics. This extends Badaskar et al. [6]’s earlier work on linguistic features by incorporating modern NLP techniques and providing empirical evidence for their effectiveness.

The gap between our hand-crafted (94% F1) and deep learning (98% F1) approaches suggests that while traditional linguistic features remain valuable, they may not capture the full complexity of modern language models. This finding advances the field by demonstrating quantitatively how deep learning approaches can better adapt to the sophisticated patterns in contemporary AI-generated text.

These results highlight an evolving challenge in the field: as language models become more sophisticated, detection methods must balance accuracy within specific domains against broader generalizability. Our work contributes to addressing this challenge by providing the comprehensive comparison of traditional and deep learning approaches across multiple model generations and identifying specific linguistic features that remain reliable across different language models. These findings suggest that future detection systems may need to adopt hybrid approaches that combine the interpretability of hand-crafted features with the adaptive capacity of deep learning models.

5.2 Linguistic patterns and model behavior

Our analysis of linguistic features revealed distinct patterns between human and machine-generated texts, corroborating and extending the findings of previous research. For example, similar to Reviriego et al. [21], who reported a narrower range of lexical diversity in ChatGPT-3.5 generated text, we observed a similar trend in both GPT-2 and ChatGPT-3.5 outputs. However, our study also examined a wider range of linguistic features, including POS distributions and

dependency relations, providing a more fine-grained analysis of the differences between human and AI-generated text.

These observed differences in POS distributions can be interpreted through the lens of linguistic theories such as Systemic Functional Linguistics (SFL), which emphasizes the role of register and genre in shaping language use [22], and Construction Grammar (CxG), which posits that language is comprised of learned pairings of form and meaning, from simple morphemes to complex syntactic structures [23]. The tendency of AI models to overuse certain POS tags, such as nouns and determiners, may reflect their reliance on statistical patterns in the training data, rather than a nuanced understanding of context and stylistic variation as described in SFL. For instance, the models might not fully grasp the subtle differences in language use between a formal academic paper and a casual blog post, leading to an over-representation of features associated with the more frequent register in their training data.

Furthermore, the reduced frequency of complex syntactic structures, as indicated by the analysis of dependency relations, aligns with theories of language production that posit a greater cognitive effort is required to generate syntactically complex sentences [24, 25]. This suggests that current AI models, while proficient at generating statistically probable sequences, may still struggle with the intricacies of human-like syntactic planning and execution, potentially due to the limitations of their underlying architecture or training methodologies [26].

5.3 Ethical implications and societal impact

The ethical implications of our research can be analyzed through various ethical frameworks. From a deontological perspective, the use of AI to generate deceptive content, such as fake news or fraudulent product reviews, violates the principle of honesty and the duty to tell the truth. Consequentially, the widespread dissemination of such content could erode public trust in institutions, damage reputations, and even incite social unrest. A consequentialist perspective would weigh the benefits of advanced language models against the potential harms, such as the spread of misinformation and the erosion of trust. If the potential for harm outweighs the benefits, then a consequentialist may argue against the unchecked proliferation of these technologies. A virtue ethics lens would focus on the impact of AI-generated text on virtues such as authenticity, trustworthiness, and intellectual integrity. For example, the proliferation of AI-generated academic papers could undermine the value of

genuine scholarly work and erode the integrity of the academic system.

The societal impact of AI-generated text is far-reaching. In the realm of journalism, the ability of AI to produce convincing news articles raises concerns about the spread of misinformation and the potential for malicious actors to manipulate public opinion. In education, the availability of AI tools that can generate essays and research papers poses a significant challenge to academic integrity and the assessment of student learning. Furthermore, the use of AI-generated text in social media and online forums could exacerbate the problem of online harassment, trolling, and the spread of hate speech.

While our study contributes to the development of more sophisticated detection methods, we acknowledge the limitations of relying solely on technological solutions. As language models continue to improve, the cat-and-mouse game between text generation and detection is likely to intensify. This highlights the need for a multi-pronged approach that includes not only technological innovation but also the development of ethical guidelines for the use of AI, media literacy education to help individuals identify AI-generated content, and potentially regulatory frameworks to address the most serious risks.

5.4 Limitations and future directions

While our study provides valuable insights, it has limitations that point to directions for future research:

1. **Dataset diversity:** Our analysis focused on specific datasets (HC3 and GPT-2), which may not fully represent the diversity of AI-generated text encountered in real-world scenarios. Future work should expand to a broader range of language models, including more recent models like GPT-4 and LLaMA, and encompass a wider variety of domains and genres to enhance the generalizability of detection methods.
2. **Model interpretability:** While the RoBERTa-based model demonstrated superior performance, its black-box nature makes it difficult to understand the specific linguistic cues it uses for detection. This lack of interpretability poses a challenge for building trust in the detection system and for identifying potential biases. Developing more explainable AI detection models, perhaps by incorporating attention mechanisms or other techniques that highlight important features, remains an important direction for future research.

3. **Trade-offs between interpretability and performance:** While our study focuses primarily on the performance of the models, the trade-offs between interpretability and performance is not fully explored here, and can be explored in future research. However, we can mention that while deep learning models like RoBERTa offer higher performance, they come with a trade-off of lower interpretability. This could be a topic of research in future work, which may consider techniques to increase the interpretability without sacrificing performance.
4. **Adversarial scenarios:** Our study did not consider adversarial attacks, where text is deliberately crafted to evade detection. Future research should explore the robustness of detection methods against such attacks, potentially by developing adversarial training techniques or incorporating methods for detecting subtle perturbations in the text.
5. **Ethical considerations:** As detection technologies become more powerful, it is crucial to consider the ethical implications of their use. For example, widespread deployment of AI text detection systems could raise privacy concerns if used to monitor individuals' communications without their consent. Furthermore, there is a risk that detection tools could be used to suppress legitimate forms of expression, particularly if they are biased against certain dialects or writing styles. Future research should address these ethical challenges by developing guidelines for the responsible use of AI detection technologies, exploring methods for mitigating bias, and engaging in broader discussions about the societal impact of these tools.

6 Conclusion

This study provides a comprehensive comparison of hand-crafted feature-based and deep learning approaches for detecting machine-generated text. Our findings demonstrate the superior performance and generalization capabilities of deep learning methods, particularly those based on large pre-trained models like RoBERTa. However, the challenges

in cross-dataset generalization highlight the ongoing need for more adaptive and robust detection strategies.

The linguistic patterns uncovered in our analysis offer valuable insights into the current state of AI text generation, revealing both the impressive capabilities and persistent limitations of language models. These findings contribute to our understanding of how AI-generated text differs from human-written content, informing future developments in both generation and detection technologies. Importantly, our work also highlights the significant ethical considerations surrounding AI text generation and detection. As AI continues to advance, the ability to reliably detect machine-generated content becomes increasingly crucial for maintaining information integrity and trust in digital communications. However, it is equally important to address the ethical challenges related to potential misuse, bias, and societal impact of these technologies.

Looking forward, the field of AI-generated text detection will likely require ongoing innovation to keep pace with rapidly evolving language models. Future research should focus on developing more generalizable and interpretable detection methods, exploring adversarial scenarios, and considering the broader societal implications of these technologies. Furthermore, a multidisciplinary approach involving not only computer scientists but also ethicists, policymakers, and the public will be essential to ensure that AI language technologies are developed and used responsibly, maximizing their benefits while mitigating potential risks.

Ultimately, addressing the challenges posed by AI-generated text will require a multifaceted approach, combining technological solutions with policy measures and public awareness. By advancing our understanding and capabilities in this domain, we can work towards harnessing the benefits of AI language technologies while mitigating potential risks and ethical concerns.

Appendix: XGBoost feature importance bar plots

See Figs. 3, 4 and 5.

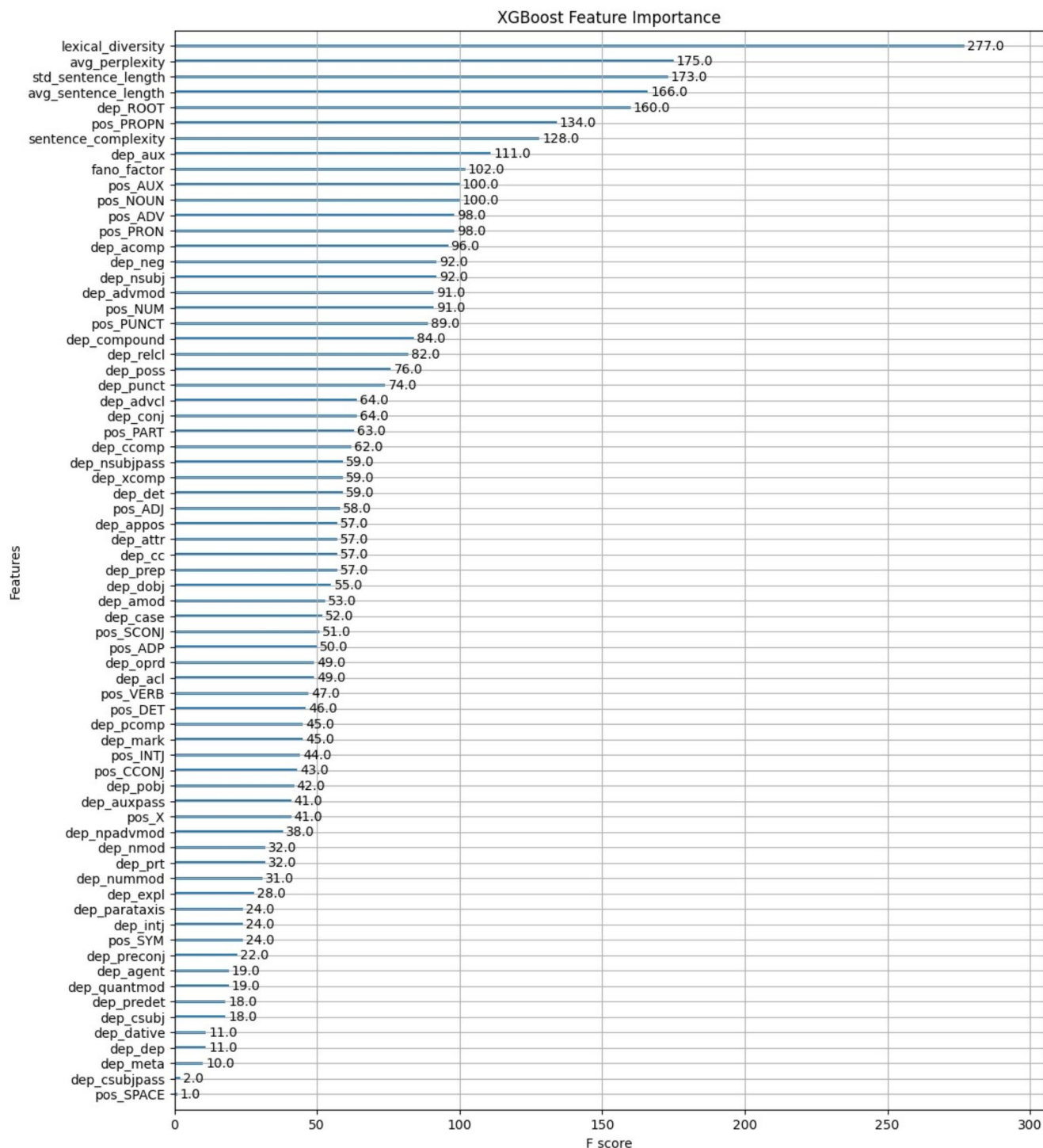


Fig. 3 Feature Importance for XGBoost Classifier trained on HC3 dataset (with indicating words). Higher values correspond to higher importance

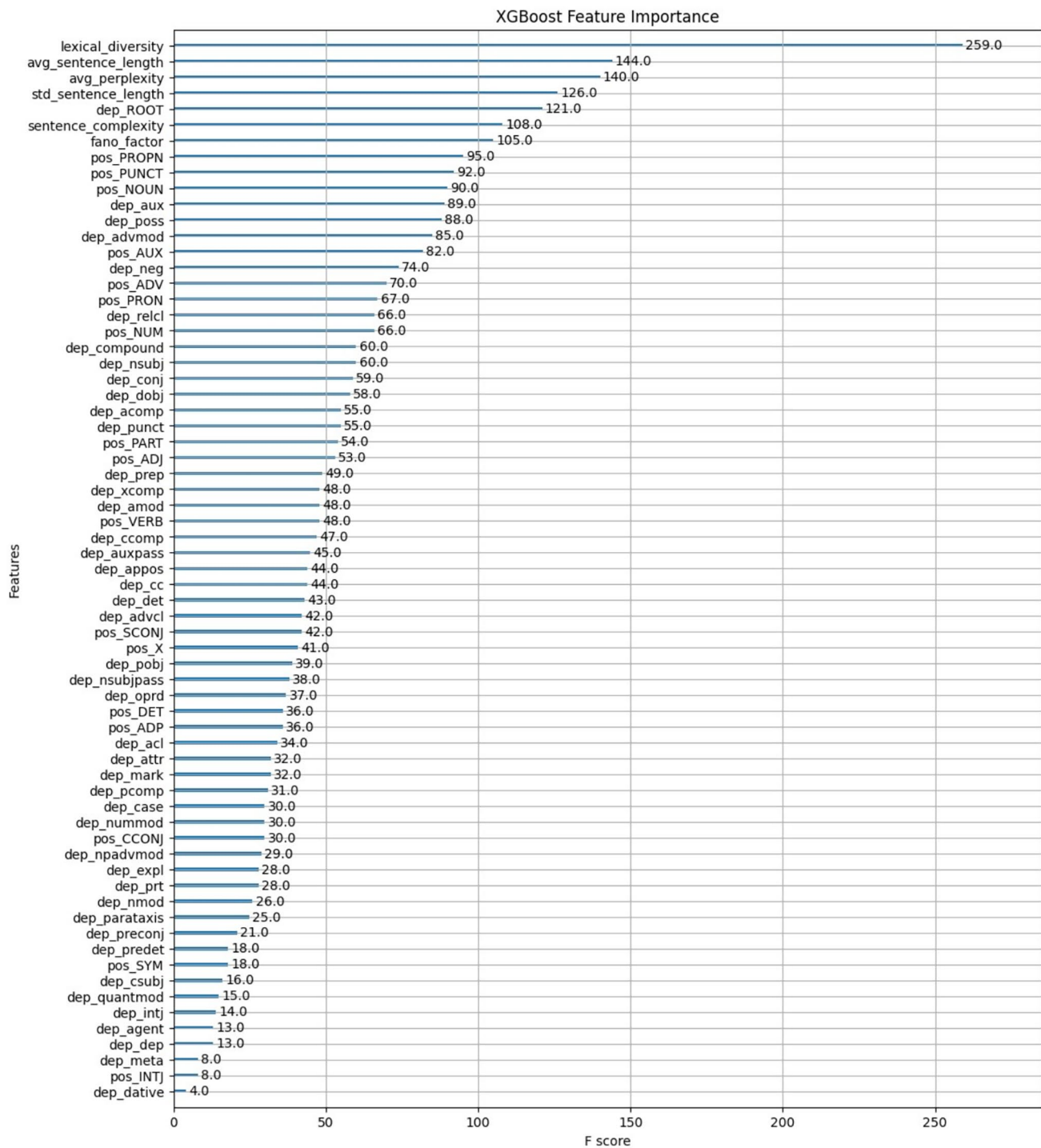


Fig. 4 Feature Importance for XGBoost Classifier trained on HC3 dataset (without indicating words). Higher values correspond to higher importance

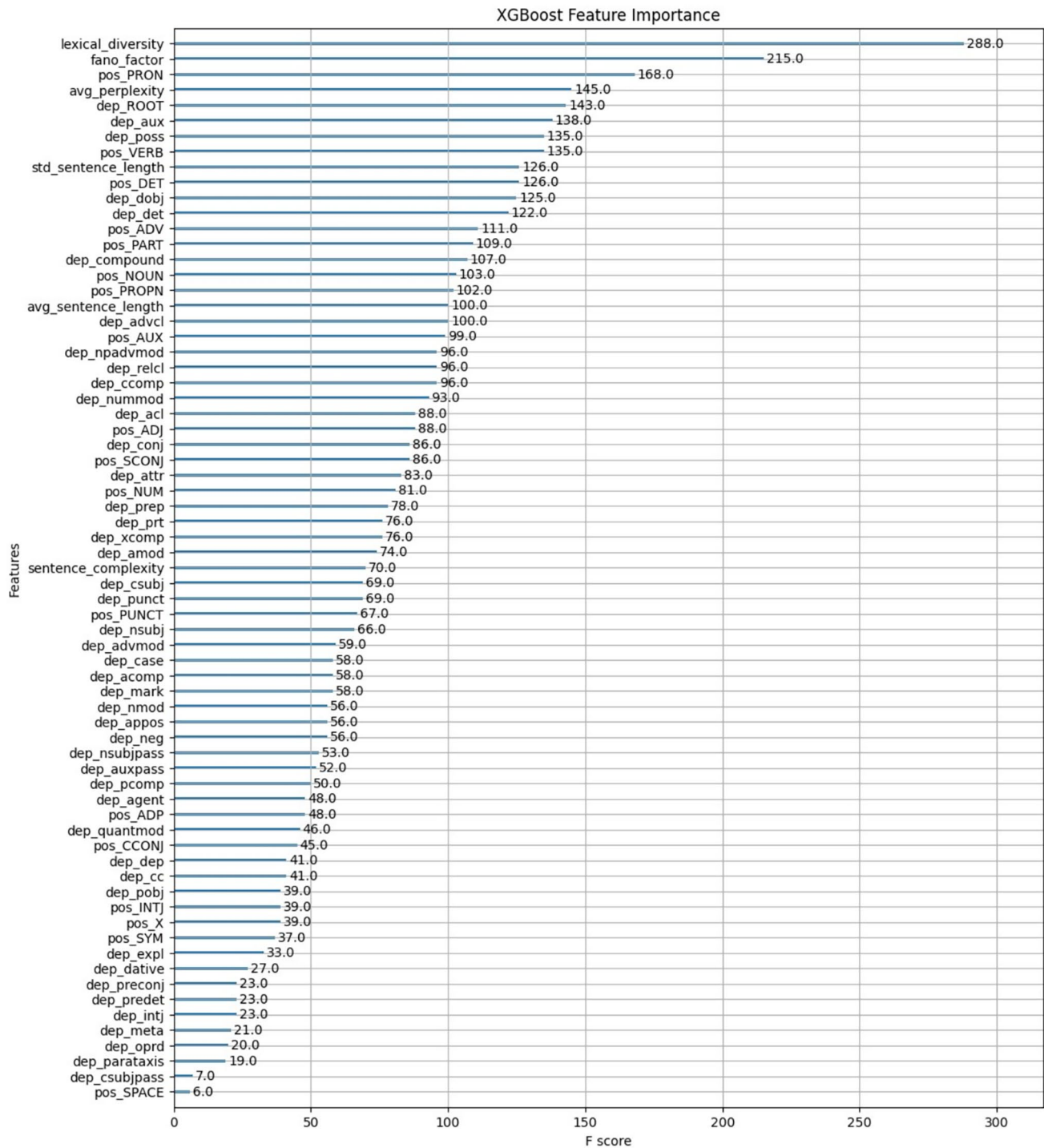


Fig. 5 Feature Importance for XGBoost Classifier trained on GPT-2 output dataset. Higher values correspond to higher importance

Conflict of interest The author declares no conflicts of interest related to this work.

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References

1. Solaiman, I., Brundage, M., Clark, J., Askill, A., Herbert-Voss, A., Wu, J., Radford, A., Krueger, G., Kim, J.W., Kreps, S., et al.: Release strategies and the social impacts of Language models (2019). arXiv preprint arXiv:1908.09203
2. Graddon, K.T.: Electric sheep on the pastures of disinformation and targeted phishing campaigns: the security implications of Chatgpt. *IEEE Secur. Priv.* **21**(3), 58–61 (2023)
3. Spitale, G., Biller-Andorno, N., Germani, F.: Ai model gpt-3 (dis) informs Us better than humans. arXiv:230111924 (2023)
4. Mohamed Firdhous, M.F., Elbreiki, W., Abdullahi, I., Sudantha, B.H., Budiarto, R.: Wormgpt: a large language model chatbot for criminals. In: 2023 24th International Arab Conference on Information Technology (ACIT), pp. 1–6 (2023). <https://doi.org/10.1109/ACIT58888.2023.10453752>
5. Erzberger, A.: WormGPT and FraudGPT– The Rise of Malicious LLMs. Trustwave SpiderLabs Blog. Accessed: 2024-09-03 (2023). <https://www.trustwave.com/en-us/resources/blogs/spiderlabs-blog/wormgpt-and-fraudgpt-the-rise-of-malicious-llms/>
6. Badaskar, S., Agarwal, S., Arora, S.: Identifying real or fake articles: towards better language modeling. In: Proceedings of the Third International Joint Conference on Natural Language Processing: Volume-II (2008)
7. Pu, J., Sarwar, Z., Abdullah, S.M., Rehman, A., Kim, Y., Bhat-tacharya, P., Javed, M., Viswanath, B.: Deepfake text detection: limitations and opportunities. In: 2023 IEEE Symposium on Security and Privacy (SP), pp. 1613–1630 IEEE (2023)
8. Guo, B., Zhang, X., Wang, Z., Jiang, M., Nie, J., Ding, Y., Yue, J., Wu, Y.: How close is Chatgpt to human experts? Comparison corpus, evaluation, and detection. (2023). arXiv preprint arXiv:2301.07597
9. OpenAI: GPT-2 Output Dataset. (2019). <https://github.com/openai/gpt-2-output-dataset>. Accessed: 2024-09-03
10. Nguyen-Son, H.-Q., Tieu, N.-D.T., Nguyen, H.H., Yamagishi, J., Zen, I.E.: Identifying computer-generated text using statistical analysis. In: 2017 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), pp. 1504–1511 IEEE (2017)
11. Zellers, R., Holtzman, A., Rashkin, H., Bisk, Y., Farhadi, A., Roesner, F., Choi, Y.: Defending against neural fake news. In: Advances in Neural Information Processing Systems, pp. 9054–9065 (2019)
12. Mitchell, E., Lee, Y., Khazatsky, A., Manning, C.D., Finn, C.: Detectgpt: Zeroshot machine-generated text detection using probability curvature. ArXiv Preprint arXiv:230111305 (2023)
13. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692 (2019)
14. Yu, X., Qi, Y., Chen, K., Chen, G., Yang, X., Zhu, P., Zhang, W., Yu, N.: Gpt paternity test: Gpt generated text detection with Gpt genetic inheritance. ArXiv Preprint arXiv:230512519 (2023)
15. Fano, U.: Ionization yield of radiations. II. the fluctuations of the number of ions. *Phys. Rev.* **72**, 26–29 (1947). <https://doi.org/10.1103/PhysRev.72.26>
16. Honnibal, M., Montani, I.: spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks and incremental parsing. To appear (2017)
17. Chen, T., Guestrin, C.: Xgboost: A scalable tree boosting system. In: Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining, pp. 785–794 (2016)
18. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E.: Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.* **12**, 2825–2830 (2011)
19. Mitrović, S., Andreoletti, D., Ayoub, O.: Chatgpt or human? Detect and explain. Explaining decisions of machine learning model for Detecting short Chatgptgenerated text. ArXiv Preprint arXiv:230113852 (2023)
20. Jawahar, G., Abdul-Mageed, M., Lakshmanan, V.L.: Automatic detection of machine generated text: A critical survey. arXiv preprint arXiv:2011.01314 (2020)
21. Reviriego, P., Conde, J., Merino-Gómez, E., Martínez, G., Hernández, J.A.: Playing with words: Comparing the vocabulary and lexical diversity of Chatgpt and humans. *Mach. Learn. Appl.*, **100602** (2024)
22. Halliday, M.A.K., Matthiessen, C.M.: Halliday’s Introduction To Functional Grammar. Routledge, London (2013)
23. Goldberg, A.E.: The nature of generalization in language (2009)
24. Levelt, W.J.: Speaking: from Intention To Articulation. MIT Press, Cambridge, MA (1993)
25. Lees, R.B.: Syntactic structures. JSTOR (1957)
26. Frank, S.L., Bod, R., Christiansen, M.H.: How hierarchical is language use? Proceedings of the Royal Society B: Biological Sciences 279(1747), 4522–4531

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